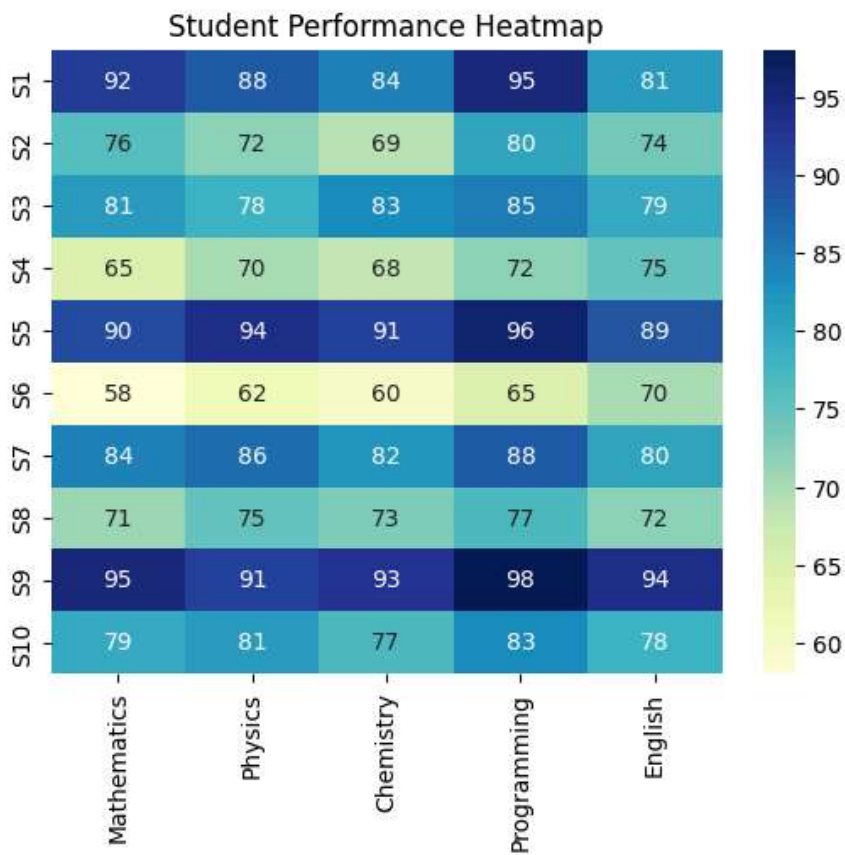


```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    'Mathematics': [92, 76, 81, 65, 90, 58, 84, 71, 95, 79],
    'Physics': [88, 72, 78, 70, 94, 62, 86, 75, 91, 81],
    'Chemistry': [84, 69, 83, 68, 91, 60, 82, 73, 93, 77],
    'Programming': [95, 80, 85, 72, 96, 65, 88, 77, 98, 83],
    'English': [81, 74, 79, 75, 89, 70, 80, 72, 94, 78]
}

df = pd.DataFrame(data,
                   index=[f'S{i}' for i in range(1,11)])

sns.heatmap(df, annot=True, cmap='YlGnBu')
plt.title('Student Performance Heatmap')
plt.show()
```



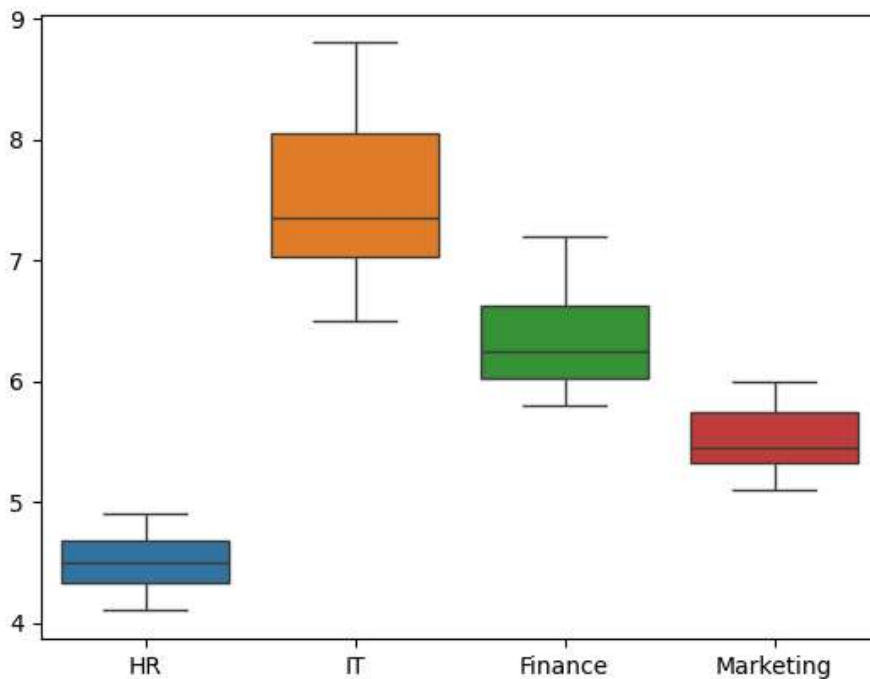
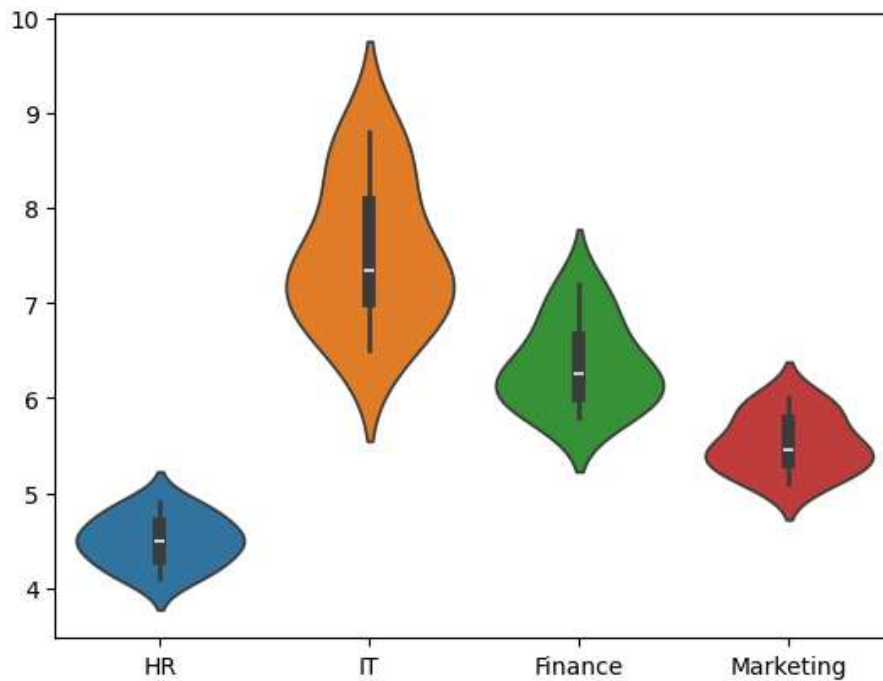
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'HR': [4.2, 4.5, 4.7, 4.1, 4.6, 4.3, 4.8, 4.4, 4.9, 4.5],
    'IT': [6.5, 7.0, 7.4, 6.8, 8.2, 7.6, 8.5, 7.1, 8.8, 7.3],
    'Finance': [5.8, 6.2, 6.0, 5.9, 6.7, 6.4, 6.9, 6.1, 7.2, 6.3],
    'Marketing': [5.1, 5.3, 5.5, 5.4, 5.8, 5.6, 5.9, 5.2, 6.0, 5.4]
})
```

```
})
```

```
sns.violinplot(data=df)  
plt.show()
```

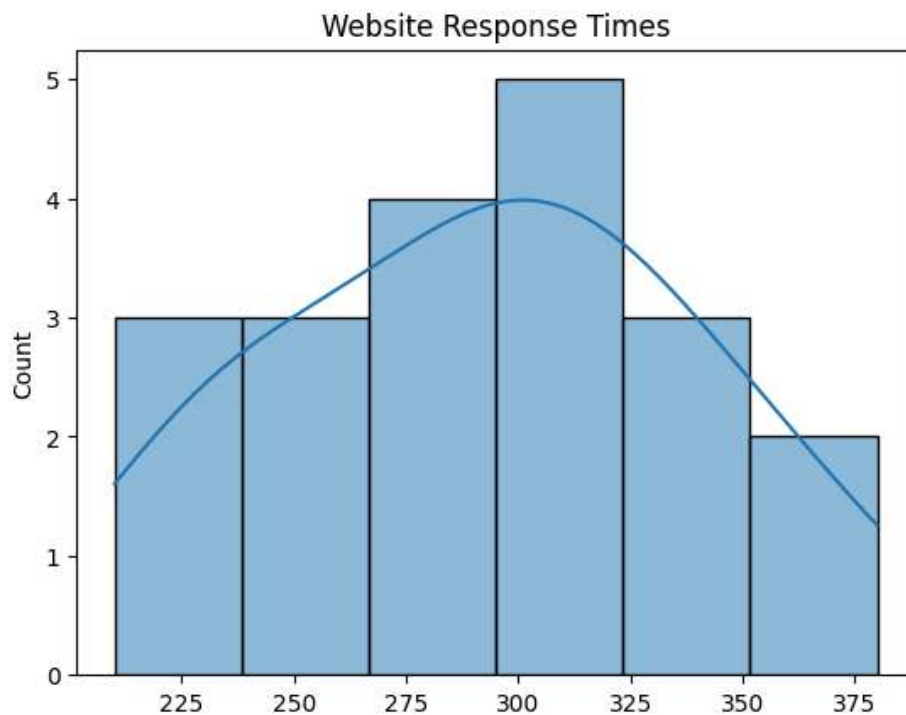
```
sns.boxplot(data=df)  
plt.show()
```



```
import seaborn as sns  
import matplotlib.pyplot as plt
```

```
times = [210,225,230,240,250,260,270,280,285,290,  
        300,305,310,315,320,330,340,350,360,380]
```

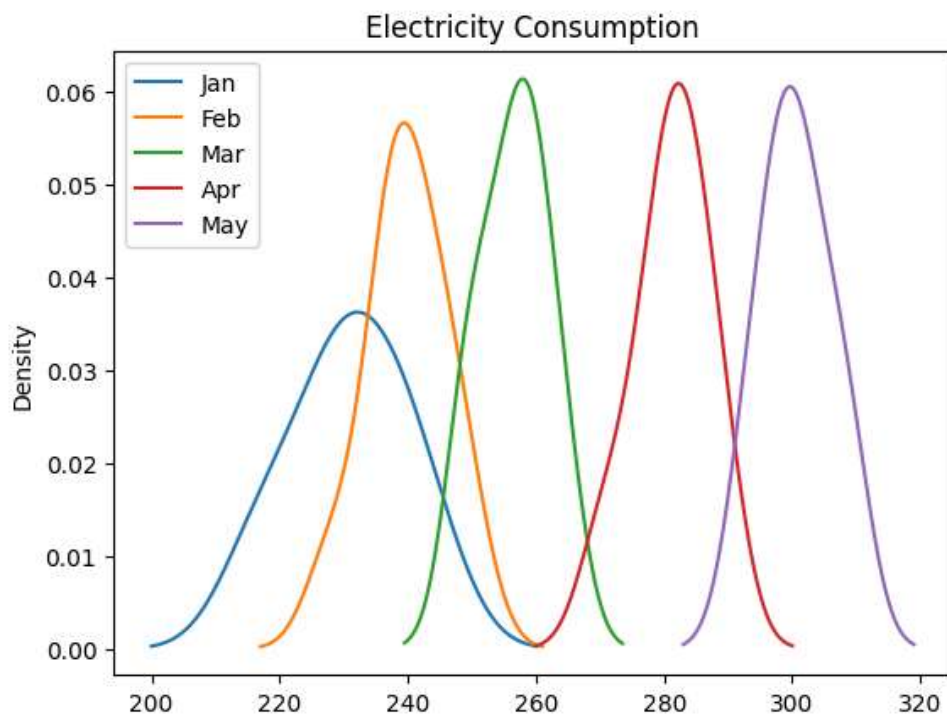
```
sns.histplot(times, kde=True)
plt.title("Website Response Times")
plt.show()
```



```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Jan': [220, 240, 230, 215, 225, 245, 235, 228, 238, 232],
    'Feb': [235, 245, 238, 228, 240, 250, 242, 236, 246, 239],
    'Mar': [250, 260, 255, 248, 258, 265, 259, 252, 262, 256],
    'Apr': [275, 285, 280, 270, 282, 290, 284, 278, 287, 281],
    'May': [295, 305, 300, 292, 298, 310, 302, 297, 307, 301]
}

pd.DataFrame(data).plot(kind='density')
plt.title("Electricity Consumption")
plt.show()
```



```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    '20-30': [9500, 10200, 9800, 11000, 10500, 9700, 10100, 10800, 11200, 9900],
    '31-45': [8200, 8600, 8400, 9000, 8900, 8500, 8700, 9100, 9200, 8600],
    '46-60': [6500, 6700, 6900, 7200, 7000, 6800, 7100, 7300, 7400, 6950]
})

df.plot(kind='hist', alpha=0.5)
plt.show()

df.plot(kind='density')
plt.show()

sns.boxplot(data=df)
plt.show()
```

